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Is Cytoplasm Intelligent Too?

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1. Introduction

Ever since modern computers arrived on the scene, the concept of “intelligence” has been greatly demystified and replaced by “data processing capacity.” Even the last bastions of mystical meaning in the word “intelligence,” namely the aspects of “creativity” and “awareness,” are being secularized. Computerists are exploring whether the use of look-up tables, tree searches, and pseudo-random number generators can replace the old-fashioned, mystical terms.

Traditionally, the genome has been considered the site of cellular information storage and processing. In the terminology of computerists, one may therefore consider the genome to be “intelligent.” By contrast, the cytoplasm is generally considered the site at which the nuclear instructions are blindly carried out. Using again the terminology of today’s computerists, it is considered as a “dumb” peripheral. Can this attitude toward the cytoplasm be maintained in view of all we have learned in recent years about the functions of the cytoplasm and the rules of engineering that govern any kind of data-processing systems?

Nobody would doubt that the cytoplasm is indispensable for a cell. After all, cells have always been made from cells and never from genomes alone (Nanney, 1968). Furthermore, it is clear that cytoplasm can carry out many functions in the absence of nuclei, as demonstrated by the motility and other actions of cytoplasts (Goldman *et al.*, 1973). On the other hand, there is little doubt that the genome has the final decision in matters of cell function. Viral replication, where the isolated viral genome can override every other cellular

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function in order to replicate the virus, demonstrates this point sufficiently. Yet, subordination under DNA instruction does not automatically make the cytoplasm a mindless slave. Even the data-processing systems of our own technology do not restrict information handling and decision making to the main computer. Today most peripherals such as printers, terminals, and disk drives have microprocessors of their own and carry out intelligent functions autonomously. The main computer would be quickly paralyzed if it had to monitor and control every contingent action of every instruction to the peripherals. Why, then, should the infinitely more complex cytoplasm be a rigidly controlled robot of the genome?

We have never encountered poor engineering in nature, and therefore one may expect that cytoplasm is an intelligent peripheral if not an almost equal partner to the genome in cellular function. Can this expectation be substantiated? Is cytoplasm intelligent, too?

If cytoplasm intelligence existed, it would be a most consequential cellular property. It would be expected to play important roles in cell behavior, cell-cell recognition, and cell communication and thus be pivotal for the understanding of development, aging, and cancer.

2. Cytoplasm: Compartments and Information

Cytoplasm is neither periodically structured like a crystal nor randomly structured like a boiling liquid, but it is an organized piece of matter. There are two minimal requirements for matter to be organized. First, it must be compartmentalized. Without compartmentalization, the system would mix its components—all engaged in quite diverse functions—to the point of chaos. The second requirement concerns its content of information.

According to Shannon's definition (Shannon, 1948), information is not concerned with the meaning of a message, but only with its formal structure. One can receive a message in the form of a spatial structure (e.g., a letter, a magnetic tape) or a temporal structure (e.g., a radio signal, a movie). The receiver of the message must decode the message and must therefore have a way of telling an intended signal from an accident of the transmission mechanism. Shannon suggested that "intended" signal be identified with "unpredictable" signal. For example, a meaningful text is a sequence of words and letters that the reader cannot anticipate. By contrast, a text that consists of an uninterrupted string of the letter E cannot carry much information. The same holds true for temporal messages, such as radio signals. The strictly periodic, and therefore predictable, carrier wave of a transmitter carries no information. Only after the periodic oscillations of the carrier wave have been modulated with unpredictable changes of frequency (FM) or amplitude (AM) can speech or music be heard.

Not only strictly periodic structures are unfit to carry information, but random structures are equally devoid of information. The reason is the same as before. A random sequence of spatial or temporal characters is predictable

by statistics. Consequently, a text consisting of an entirely random sequence of letters is meaningless, just as a broadcast with entirely random modulations (i.e., with so-called white noise) cannot transmit any messages.

Shannon developed a method of quantitating information based on the degree of unexpectability of the next character or event in the message-carrying structure. Ever since, information can be treated as a quantity and measured in amounts.

Returning to the second requirement of organized matter, it must contain a certain amount of information. Without it, no component would find its compartment. It is also obvious, from this requirement, that cytoplasm must not be structured in a strictly periodic manner like a crystal, nor must it be random like a boiling liquid. In both cases, it could not carry any information.

Clearly, cytoplasm fulfills these requirements. It contains numerous compartments, such as organelles, polymer arrays, and membranes, and it contains vast amounts of information in the form of specific molecules and their nonrandom arrangements. None of the compartments is ever strictly periodic, either. Ironically, the necessarily noncrystalline structure of the cytoplasm had for many years been the source of the widespread belief that it is randomly structured.

The presence of compartments and information alone cannot distinguish cytoplasm from a filing cabinet. For a more realistic description of the living cell, the cellular information and compartments should be considered to be in motion. As the borderlines of compartments begin to shift, their actions change in character and intensity; contacts with new neighbors are made; some interactions are eliminated and others are initiated. The traffic patterns of energy and molecules change, and information is exchanged. During the course of uncountable simultaneous biochemical and biophysical events, information is copied, received, transmitted, and stored as hormones, mRNAs, neurotransmitters, pre- and postsynaptic potentials, post-translational modifications, covalent links between macromolecules. Different kinds of cyclic events keep time and synchronize other cellular events, eventually generating mitotic cycles and circadian rhythms. Materials such as proteins and RNAs or much larger structures such as viruses or organelles that carry the information around are transported, produced, exchanged, recycled, modified, inhibited, or stored.

All these events occur in a microscopically small world of violent and random thermal fluctuations. Yet, cytoplasm can keep its complex actions accurate in the midst of drowning thermal noise all around and within.

3. Cytoplasm as a Random Machine

What, then, is cytoplasm? In a first approximation, one may seek the secret of the cytoplasmic complexity in the very randomness of the thermal chaos combined with the experimentally observed specificity of biochemical

reactions. If the fact is added that cytoplasm contains very large numbers of molecules, then one may arrive at a

First Approximation of Cytoplasm: Cytoplasm is a giant set of molecules with high specificities in free competition with each other.

This cytosol concept of cytoplasm emphasizes the necessity to purify and characterize specific biological molecules but ascribes little significance to their spatial distribution within the cell. Yet, electron microscopes demonstrate the extraordinary complexity of the spatial arrangements of the cytoplasm. Immunocytochemistry also shows quite convincingly that cytoplasm is not a jumble of specific molecules bouncing around at the whim of thermal fluctuations. On the contrary, many proteins, such as actin, α -actinin, myosin, tropomyosin, are found arranged in elaborate nonrandom spatial patterns.

4. Cytoplasm as an Automatic Machine

In view of these patterns, new mechanisms of control of chemical reactions become conceivable, that are based on supramolecular topology and architecture, rather than on nothing else but freely swarming inhibitors and promoters, with their competing binding constants. With these spatial arrangements of molecules, it becomes likely that there is cooperativity between biochemical events over large intracellular distances. Biochemical reactions in different parts of a cell may become coordinated in a way similar to that of multienzyme complexes. Thus, the increasing recognition of cytoarchitecture may give rise to the

Second approximation of cytoplasm: Cytoplasm is a giant multienzyme complex.

This cytoarchitecture concept emphasizes the fibrous, polymeric configurations of cytoplasmic proteins, such as microfilaments, intermediate filaments, and microtubules. They seem able to express aspects of cooperativity (e.g., flagellar movement) and to serve as frameworks for associated, enzymatically active cytoplasmic proteins (e.g., metabolic enzymes).

With the second approximation, we have effectively shifted the emphasis from “random” to “automatic” in an attempt to characterize the principle of cytoplasmic actions. Both principles mark fundamentally different attitudes toward biology and entail quite different approaches to the question of life’s origin. I submit that there is yet another step to be taken along this way, namely, the acceptance that cytoplasm is “intelligent.”

5. Cytoplasm as an Intelligent Machine

Intelligent objects are able to collect data, process them, and make decisions on the basis of these data and of built-in criteria that distinguish between

“desirable” and “undesirable” outcomes. They are able to assess global situations around them and do not merely react to local stimuli whenever they occur. They communicate data with other intelligent objects and adjust their actions accordingly.

If we observe cytoplasm expressing such properties, we may move somewhat away from the concept of a chemical automaton with its thousands of intertwined chain reactions and look more seriously into the possibility of a chemistry-based, gelled, or even liquid data-processing system.

The idea of computerization of a machine is new and exciting for today's human technology. Yet, like so many other new ideas of human technology, this one might also have been in use by nature in the cytoplasm of cells for millions of years. Unfortunately, we cannot look into our own computer technology for easy answers and analogies. If cytoplasm can be considered computerized, it would be a liquid or gelled computer throughout. Although there are so-called “liquid computers,” they are made of solid pipes with liquid flowing inside. There is no precedent for a truly liquid computer in human technology. To my knowledge, there is not even a clue as to how to build a liquid miniature computer that would function despite thermal fluctuations and other turbulences in the liquid that would disrupt the circuitry. Nevertheless, this may be a major challenge of modern cell biology, namely to combine biochemistry with computer science and microtechnology. Perhaps, cells can teach us to construct truly liquid computers that are even able to reproduce themselves.

In order to meet this challenge, we would have to place greater emphasis on macromolecules as carriers of cellular information and less emphasis on their properties as chemicals. In the terminology of today's computers, the molecular turnover would acquire an aspect of memory-refresh operation; post-translational modifications of macromolecules would suggest aspects of labeling and flag setting; the cellular environment (i.e., the place from which information-bearing molecules are retrieved irrespective of their source) would become comparable with a data bus; cellular clocks would be considered timers for the coordination of information fluxes; not only DNA and RNA, but other heteropolymers as well would be considered candidates for coded information storage; macromolecules with multiple binding sites (e.g., fibronectin) could be code carriers, too, depending on the pattern of occupation of these sites; proteins that decorate the length of microfilaments, intermediate filaments, or microtubules would be suspected of being potential elements of a code; the spatial pattern of crosslinking proteins of fibrous structures would not be viewed as merely affinity-driven and randomly placed, but as a most complex system of latches or switches that may divert or interrupt the flow of information involving the fibers; shock responses of cells may be seen in the light of interrupt routines, so essential in human computer programs; and seemingly useless cell components might acquire a most important role in power-up routines after an interruption, the action of which may prevent the data-processing system from “getting hung up.”

In human data-processing technology, there is a sharp division between

changeable fast-access memory (random-access memory, RAM), unchangeable fast-access memory (read-only memory, ROM), unchangeable slow-access memory (ticker tapes, laser disks, bar codes), and changeable slow-access memory (tapes, disks). If only we look for them, we may find their counterparts not only in the nucleus, but in the cytoplasm as well. Sequestered calcium, cyclic adenosine monophosphate (cAMP) concentrations and distributions, positions of mitochondria or adhesion sites, orientation of centrioles, distribution of nuclear pores, and many other changeable structural features of cytoplasm and nucleus may reveal aspects of "memory." After all, at every cell division each daughter receives a huge amount of nongenomically stored information through its half of the maternal cytoplasm. We do not know yet how volatile or how permanently stored this cytoplasmic information is.

With these concepts in mind, we may suspect and explore relationships between cellular functions and compartments implied by data processing rather than by chemistry alone. Most excitingly, I think, we may some day even go beyond this concept and begin to explore whether cells are able to learn. Experiments about cellular learning are presumably best carried out with protozoa exposed to artificial environments with obstacles and substances they have never before encountered. To my knowledge, they have not yet been tried.

In the light of such an interpretation, our wealth of chemistry-based knowledge would not become obsolete. On the contrary, most biochemical data would become upgraded. Instead of placing them within the context of the widespread belief that "nothing goes on in cells but thermodynamics," they might acquire additional meaning and weight if we were to become convinced that "at least one other thing goes on in cells, namely data processing." With such a change of attitude, we would arrive at a

Third approximation of cytoplasm: Cytoplasm is an intelligent, giant multienzyme complex.

To my knowledge, cytoplasm intelligence has been studied in three different cases. One case concerns the alternation between tumbling and smooth swimming of motile bacteria in chemotactic environments (Adler, 1969; Macnab and Koshland, 1972). The second case concerns the backward swimming of the ciliate *Paramecium aurelia* after collision with an object (Kung *et al.*, 1975). The third system is the amoeboid locomotion of cultured 3T3 mouse cells—the topic of this chapter.

It is not surprising that all three cases have to do with cell movement. Since we do not expect a single transistor inside a computer to express artificial intelligence, we should not expect a single macromolecule inside the cytoplasm to express cytoplasm intelligence. Rather, such intelligence requires the interactions of vast numbers of molecules in complex schemes of interaction that leave little to chance. Therefore, the experimental demonstration of cytoplasm intelligence must leave the cytoplasm intact. Furthermore, it should demonstrate the cell's reactions to global situations without involving long-term genomic actions. Both criteria apply to experiments with

cell migration: A moving cell is necessarily acting as a whole and is interacting moment by moment with an unpredictable environment, which is a global situation in the first place. This chapter focuses on the question of whether the eukaryotic cell cytoplasm is intelligent. The work on prokaryotes and ciliates is summarized elsewhere (Koshland *et al.*, 1976; Kung, 1976; Springer *et al.*, 1979).

6. A Brief Attempt to Place the Topic in Perspective

There are eight major categories of eukaryotic cell expression:

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|----------------------|-----------------------------|
| 1. Replication | 5. Macromolecular synthesis |
| 2. Differentiation | 6. Communication |
| 3. Transport | 7. Repair and maintenance |
| 4. Energy Metabolism | 8. Movement |

We restrict our topic by focusing on the last category, movement, which in turn can be divided into three major categories: intracellular movement, surface movement, and locomotion. Cell locomotion occurs by movement of cilia or flagellae, by cell body deformation (amoeboid motion), or by other methods (e.g., parasitism or symbiosis; cf. Cleveland and Grimstone, 1963).

We restrict our topic further by choosing to discuss the amoeboid form of locomotion. This expression of motility needs to be studied in two subcategories: (1) the mechanism (how does the movement come about?), and (2) its control (why does the cell migrate in this direction and not another?).

The mechanism of amoeboid cell migration has three major aspects that are studied by today's cell biologists:

1. Potential mechanism of force generation (e.g., contraction of an actomyosin system, assembly–disassembly of fibrous structures, dynein–tubulin sliding, filament rotation)
2. Transmission of motile forces to the framework of a cell (cytoskeleton in general, anchorage of fibrous elements in the cell surface in particular)
3. Generation of traction (adhesion plaques, extracellular matrix, basement membranes)

The control of cell migration involves three subtopics as well:

1. Generation of a front-rear axis (polarity of a cell, role of microtubules in establishing the front of a cell)
2. Cell body coordination (How do different parts of a cell cooperate in order to displace its body?)
3. The logic of migration (Are there rules of changing direction? Can migrating cells make decisions? Can migrating cells assess their environment globally?)

After all this dividing, subdividing, and sub-subdividing of topics, we have arrived at a tiny facet of cellular expression, namely the control of amoeboid cell migration, that I think provides evidence of the intelligence of cytoplasm. There are two major reasons for this expectation. First, amoeboid motion with its complex body deformations requires coordination of many domains of the cell body. Hence, the cell can be expected to process information about the metabolic and motile state of these domains. Second, locomotion requires some kind of navigation if the cell migrates into a direction that it has not taken before. Navigation, in turn, requires assessment of the entire environment and, therefore, collection and integration of data about the environment. These data may concern topological properties such as curvatures, or chemical properties such as the substrate coating, or signals such as hormones and the presence of other cells.

7. Cell Body Coordination: The Existence of Microplasts

The most immediate “world” a moving tissue cell has to assess globally and to handle is its own deformable body. There are about 40 billion protein molecules in the cytoplasm. Eventually, all these molecules must be kept under cellular control. We can expect that nature resolved this problem with the same method used by any other organization, namely by hierarchical stratification. Each stratum of organization consists of various cooperating compartments that control the next lower stratum in the hierarchy. It is therefore important to demonstrate that cooperating compartments can be isolated from cytoplasm. More importantly, one has to demonstrate their autonomy with respect to one or the other biological function. Compartments within a stratum of organization are legitimate only if they are able to carry out at least one task by themselves, without requiring anything from the outside but energy, raw materials, and the trigger signals to initiate their function. A compartment may be autonomous with respect to a particular function and dependent with respect to another. For example, mitochondria are autonomous with respect to oxidative phosphorylation, but they depend on the cellular nucleus for the synthesis of many mitochondrial proteins, such as cytochrome oxidase.

Several autonomous cellular compartments are known. We mentioned mitochondria and nuclei. With respect to migration cytoplasts (enucleated cells) are autonomous compartments as well (Goldman *et al.*, 1973). The surface of ciliates such as *Paramecium aurelia* and *Tetrahymena pyriformis* contains rows of cilia arranged in units that replicate autonomously (Sonneborn, 1970; Ng and Frankel, 1977). Isolated myofibrils of striated muscle cells are able to contract without the rest of the cell. Are there such independent units of amoeboid cell motility as well? The best-known portions of autonomously motile cytoplasm are blood platelets, which are fragments pinched off from megakaryocytes. They can produce filopodia and tiny lamellipodia (hyalomer) that ruffle occasionally (Allen *et al.*, 1979). The growth cone of a

neuron can move autonomously: the cutoff axon will continue to produce lamellipodia and filopodia (Shaw and Bray, 1977). Recently, I found that tiny fragments (2% of cell volume) can be torn out of living fibroblasts (Albrecht-Buehler, 1980). The procedure is quite simple. The cells are exposed to cytochalasin B until they arborize. Subsequently, the cells are washed away, leaving only tiny attachment fragments behind. These reseal and can be removed from the substrate by trypsinization. Although originally associated with immobilized attachment plaques, they now begin to move filopodia, ruffle lamellipodia, or produce blebs. Some fragments are so small that one may consider them isolated active filopodia, ruffles, or blebs. Whatever type of motility they express, they repeat it stereotypically for 8 or more hours, until they die. I have suggested that these fragments be called "microplasts."

Interestingly, they cannot migrate. By contrast, whole migrating cells and cytoplasts express coordinated ruffling at the front edge and form a retracting tail on its opposite side. Both the ruffling leading edge and the tail are much larger than one microplast, and their location on the cell body changes when the cell makes a turn.

Locomoting cells sometimes leave cytoplasmic fragments behind. The fragments have a "frozen-in" polarity in the form of the parallel arranged microfilaments and microtubules, the result of gross stretching before the cell broke the connection and moved away. Without the rounding and resetting effect of trypsinization used in my method to produce microplasts, the torn-off fragments retain their last polarity and may even migrate, although in a very stereotypic way as well.

The existence of microplasts suggests that cells contain cytoplasmic domains capable of autonomous amoeboid movement. If so, they must be controlled by a superior stratum of organization in the cell. Otherwise, incoherent movement of different parts of the cell body would result, making directional migration altogether impossible. This stratum of control is not located in the nucleus, because enucleated cells show normal directional migration behavior.

8. Do Filopodia Probe the Way?

It is known that cells are sensitive to light in a wide range of wavelengths. Yet, obviously, they cannot "see" optical images. Light with a wavelength short enough to be used by a tiny hypothetical cellular "eye" would have to be in the X-ray range. There are no known refractive media for X rays to make lenses; also, X rays would destroy the "eye." Yet, theoretically cells could feel out their environment. Do they?

Freshly plated cells that have just settled on a new substrate produce filopodia that wave about and that may become attached to the substrate or to another cell. If freshly plated 3T3 mouse cells settle on a glass substrate next to a tiny island consisting of a thin film of gold, the waving filopodia may touch the gold island. I found that the filopodia attached to the island and

remained there until 2–3 min later, when the cell began to produce a lamellipodium that flowed along the filopodium toward the gold. Eventually, the whole cell moved onto the island. The lamellipodium was not drawn out of the tip of the filopodium by a certain wetting force. It extended from the base of the filopodium, as if responding to a signal from the attached tip of the filopodium. Therefore, I have suggested that filopodia might serve as the organs used by cells to probe their environment (Albrecht-Buehler, 1976). The filopodia may not be the only surface projections that probe, and they may have additional cellular functions. Yet, if they are probing organs, which stratum of organization receives and processes the signals from their tips? What are the signals?

There is reason to believe that the signals involve the isometric contraction of the attached filopodium: If the gold touched by the filopodium is not tightly bonded to the glass substrate as in the above experiment, but forms a layer of loosely attached particles, the filopodium invariably hauls them toward the cell body upon contact.

9. Symmetrical Sister Cells—Predetermined Pathways?

Consider a tissue cell with “probing” filopodia migrating across a glass surface covered with loosely attached gold particles. Not only the filopodia in front, but the entire leading front as well were found to pick up the particles in front of the cell. In this way, the cell clears the particles out of its way, leaving a particle-free track behind. I suggested that such tracks be called “phagokinetic tracks.” Whenever a cell divides, its track branches into the two tracks of the daughter cells.

It was rather surprising to find that 60% of branching tracks of sister 3T3 mouse cells were geometrically related to each other, either as mirror images or as identical copies. Not only the migrational path but even the complex patterns of actin-containing microfilament bundles inside the two sisters were related to each other at the same percentages (Albrecht-Buehler, 1977). The patterns of neurites in sister neuroblastoma cells were found to express a similar geometric relationship (Solomon, 1979).

Consequently, it appears that in addition to an organizational stratum which coordinates the bodies of migrating cells, there also seems to be a level of organization that contains long-term instructions—perhaps contained in symmetrically or identically structured cell bodies—for the future itinerary of the migrating sisters. What happens if the program cannot be carried out? What happens if a 3T3 cell collides with another? Again, the phagokinetic tracks revealed a surprising logic. After colliding, the cells moved away from each other at the approximate angle of collision: The collision tracks of cells looked like the tracks of colliding billiard balls. Of course, there is no elastic collision between cells. The many complex events that take place during cell–cell collision that last for hours and include phases of “recognition;” cytoplasmic reorganization, and contact retraction are far too slow for any

measurable momentum and kinetic energy. Rather, a geometric rule of re-orientation after collision seems to be programmed into 3T3 cells and is expressed in the tracks.

10. Contact Guidance—Cell Decisions at the Crossroads?

One of the truly intriguing expressions of cell body control is the ability of cells to follow prescribed paths on their substrate. These paths may be scratches, ridges, substratum-attached fibers, lines of different materials, and so forth. Paul Weiss, who discovered this phenomenon, named it “contact guidance” (Weiss, 1961). Since then, numerous explanations ranging from differential wetting forces to contact-triggered microfilament assembly have been suggested. Whatever the particular hypothesis, they all assume that the cell has no choice but to follow the physicochemical influence of the pathway on the substrate. What would it mean if this assumption is wrong and the cell has a choice? If it follows such a path voluntarily, would this not cast a new light on its ability to assess its environment and to make decisions about it?

In order to test the cell’s ability to choose between the pursuit of a prescribed path on the substrate and ignoring it, I offered 3T3 mouse cells a particular substratum (Albrecht-Buehler, 1979a). It consisted of a 300-Å thin film of evaporated gold on a glass surface. Thin “roads” were scratched into the gold film, consisting of a stretch of exposed glass surface underneath the gold. 3T3 cells can migrate on the continuous gold film and on plain glass. Furthermore, the “roads,” with their depth of only 300 Å, could not possibly present a mechanical obstacle to the cells that lift their leading ruffles 100 times higher than that. Indeed, after following them accurately for an average of $\sim 300\ \mu\text{m}$, I found that the cells left the guiding lines.

This observation suggested that the guided cells had a choice all along. Why, then, did they follow the lines? The answer is not known. I suspect that the cells mistake the lines for guiding structures in the embryo that they are programmed to follow. Whatever the explanation, the cells can assess a topological feature of their environment regardless of the chemical composition of the guiding structure (Curtis and Varde, 1964).

Observation of guided cells at higher magnification showed that they extended their bodies far beyond the side of the glass road into the adjacent gold areas. This finding further supported the notion that the guiding influence was not overpowering for the cell in the sense of confining their bodies to the exposed glass surface of the road. Whatever differential influences the contacts with gold or glass may exert on the cell surface, both coexisted in variable and often equally large portions of the cell surface.

The aspect of choice during contact guidance was even more impressive on substrates with a network of identical roads that intersected each other in numerous places. Guided cells arriving at such crossroads were found to extend pseudopodia tentatively into each optional direction of the crossing roads before turning into one of them. This probing was expressed in pha-

gokinetic tracks as tiny “thorns” sticking out of the tracks in the direction of the crossing roads, where phagokinetic tracks came to an intersection. Why did the guided cells not simply lose guidance at the intersection and “walk away” in some random direction? After all, they could migrate on gold and glass as well.

The onset of probing at intersections suggests that the cells have a way to tell roads from intersections. In other words, they distinguish between two different spatial distributions of gold and glass. It seems possible that the cells simply react to the relative amounts of gold and glass. Obviously, an intersection has more glass per unit area than does a road. However, the migrating cells continuously experienced large changes of exposure to gold or glass while following a guiding line. It seems that as they recognize their encounter with an intersection, the cells are able to determine the spatial distribution of gold and glass on the surface of the substrate. This task seems to require data collection and processing.

11. Group Migration—Communication about Direction?

During important phases of embryonic development of animals (e.g., limb formation), entire populations of cells migrate together. Theoretically, collective migration may be the result of (1) the same exogenous driving force (e.g., chemotactic or haptotactic gradient) acting the same way on all the individuals, pulling each separately; (2) a leader cell dragging the others physically along; or (3) complex ways of communication between individuals as we know it from collective movements of organisms (e.g., a shoaling of fish).

In order to distinguish among these possibilities experimentally, one must be able to manipulate the cellular environment and observe the subsequent changes of the population behavior. Unfortunately, this is quite difficult to do inside an embryo. In tissue culture, however, the cellular environment can easily be manipulated. If it is possible at all to observe group migration of cells in culture, the distinction among the above alternatives can be made for these cells. Single PtK1 cells (cultured epithelial cells from the kidney of the marsupial, *Potorous tridactylis*) migrate very little. I recently found, however, that they can migrate directionally in groups of up to 50 cells. The cells of a group contact each other without fusing into a syncytium (Albrecht-Buehler, 1979b).

If one attempts to explain this behavior, it is obvious that a chemotactic gradient cannot be the cause of the concerted migration. No such gradient lasts for more than a few minutes in the liquid medium, whereas the groups migrate for days at a time in one direction. It cannot be caused by a haptotactic gradient on the substrate, either, because the test substrates are cleaned and covered randomly with gold particles for the phagokinetic track assay. It cannot be the result of a tug of war between the individual cells, because very few of them migrate as individuals in the first place. A tug of war would slow down each individual, because the others are countering its actions. In con-

trast, the groups were often faster than individuals. The possibility of a single leader cell dragging all others along by force becomes very unlikely if we observe migrating groups alive at high magnification. Here we find different cells in a group carrying out movements that different parts of single migrating cells used to perform. There are cells forming one giant leading ruffling lamellipodium, while others deform themselves into tails. Obviously, the cells are not dragged, but cooperate. The only explanation left is that cell communication within the group establishes a common spatial direction of displacement.

We have come full circle and arrived at a higher stratum of cellular organization. Instead of microplasts coordinating their potentially autonomous movements within a single migrating cell, there are now single autonomous cells that integrate into one migrating group. I submit that neither case can work by local biochemical reactions or by long-term genomic programs alone, but requires fast responsive data-processing systems, i.e., cytoplasm intelligence.

12. Centrioles and Cell Polarity

Which cellular organs, structures, and mechanisms could be the carrier of cytoplasm intelligence? We suspect that they would be found to be involved in the control of cell migration. This means that they are involved in the determination of cell polarity. Whatever mechanism determines the location of the next leading front of a cell is also the mechanism of migration control. Control of locomotion is equivalent to control of cell polarity. Therefore, the mechanism of cell polarity is likely to guide us to the mechanism of cytoplasm intelligence.

From numerous experiments with microtubule-disrupting drugs we know that the intactness of the microtubular network is a prerequisite for cell polarity (Vasiliev and Gelfand, 1976). In turn, the microtubules are organized by so-called microtubule-organizing centers (MTOCs). MTOCs are present at the spindle poles of all higher plant cells and of yeast, both of which do not migrate. In migrating animal cells, we find one or two additional and extraordinarily symmetrical structures associated with the MTOCs for the interphase intracellular microtubules, for the mitotic spindle microtubules as well as for the ciliary and flagellar microtubules. These are the centrioles, as well as the identically shaped basal bodies of cilia and flagella. It should be emphasized that the above-mentioned microtubule-disrupting drugs do not disassemble the centrioles.

The enigma of these remarkably structured cell parts has challenged cell biologists ever since their discovery. Originally, it was the relationship between the pair of centrioles, the so-called diplosome or cytocenter and its cyclic transformation into spindle poles, that pointed to a fundamental role of centrioles in cells. Later on, the electron microscope demonstrated the elaborate and exotic ultrastructure of centrioles and basal bodies and heightened

the enigma even more. Perhaps, most remarkable next to its unparalleled structure is the degree to which nature has conserved the centriole structure. Disregarding a few exceptional basal bodies of insect sperms and the slightly modified centrioles of nematodes, one can say that every cell with centrioles, be it a protozoon or a human tissue cell, has the same structure of the centriole. What is the function of centrioles that seems to be so basic that nature conserved it so well?

For a while, it was believed that centrioles organize mitotic spindles. However, higher plant cells form functional mitotic spindles without having centrioles at all. In particular, the diatoms demonstrated within the same organism that there is no need for centrioles in mitotic spindles. During all their mitoses, diatoms do not contain centrioles, but in meioses they assemble them *de novo* to serve later as basal bodies for the sperm cells (Manton *et al.*, 1969).

There is a striking discrepancy between the ubiquity and conservation of centrioles on one hand and their apparent uselessness on the other. For example, animal cells place a pair of centrioles at each pole of the mitotic spindle. Yet, mitotic spindles can function without centrioles even in animal cells (Berns and Richardson, 1977). Flagella carry basal bodies at their base, yet they are able to beat without basal bodies (Satir, 1965). Combining various microscopic techniques with the method of phagokinetic tracks, I found that one of the two centrioles of 3T3 cells was oriented along the direction of migration and the other perpendicular to the substrate (Albrecht-Buehler and Bushnell, 1979). Yet, displacement of a cell does not require centrioles, as demonstrated by the migration of various amoebae. Another case can be found in the sensory processes of sensory cells. Retinal rods and cones, olfactory flagella, and kinocilia in the inner ear grow out of the centriole at their base. Yet, the actual receptors for the various exogenous signals (the retinal disks, the chemoreceptors of olfactory flagella, or the microvilli surrounding the kinocilium) are not the centrioles.

In short, wherever centrioles are associated with a cellular function, they seem unnecessary for the function itself. Why, then, should nature attach such an elaborate, unparalleled, and highly conserved structure to this function? How could such widely different functions (e.g., mitosis, ciliary beat, light reception) share the common involvement of a centriole? Do these functions have a common denominator, or is the centriole just a “selfish” or “junk” cell organelle?

I believe there is a common denominator, namely the spatiotemporal integration of these functions. Whether it is the modulation of flagellar beat that steers the cell toward its goal, the tuning of the metachronal waves of cilia that propel the cell body or fluid in the proper direction, the body of a sensory cell that has to be adjusted to varying conditions of direction and intensity of the signal source, the separation of sister chromatids that has to be in tune with the segregation of all other duplicated cell structures, or the polarity of an amoeboid migrating cell inside an organism that needs to be oriented relative to the substrate and obstacles, each case seems to require the integration of various influences over the entire cell body. In this context, it is impor-

tant to note that enucleated cells that retain their ability to migrate directionally (Goldman *et al.*, 1973) also contain the centrioles (Shay *et al.*, 1975).

I believe the centriole is a key element in this cellular mechanism of integration in that it measures where in space the events occur that are relevant to any of the mentioned functions.

13. Does the Structure of Centrioles Imply Their Function?

Centrioles and basal bodies are cylinders with nine concave blades along the mantle. Each blade slants outward at an angle of 30° – 45° relative to the tangent. In cross section, each blade consists of a triplet of microtubules, two incomplete C-shaped ones and a complete O-shaped one, that are fused longitudinally (see Fig. 1). There is a peculiar pitch to each blade. As it runs down the mantle of the cylinder, it gradually advances to a final increment of one-ninth of the perimeter, just below the position of the next blade. Another remarkable geometric property of centrioles can be seen in cross section. If one draws a line through any blade, this line intersects with the inner edge of the preceding blade.

Assume that we have to integrate many unpredictable events inside and around a cell. First, we would need a specific signal emitted by each such event. Second, the signals must be located in a particular measuring device, that detects intensity and direction from where it comes; otherwise, the mechanism could not distinguish two events in different places from a single (stronger) event in one place. Third, the unknown signals must travel along straight lines; otherwise, it would be useless to determine the direction from which they happen to arrive finally at the detector.

Very few candidates would fulfill these requirements for a hypothetical signal in cellular dimensions. An example of a highly specific yet ubiquitous and straight-line propagated signal in the natural cellular environment would be the infrared radiation of the molecules inside and around cells.

Yet, whatever the signal may be, the design of an instrument to locate its direction is surprisingly determined, if one postulates that it is optimal like so many other designs of nature. One expects simplicity of form, simultaneous alertness for all directions, minimal size of blind spots, and optimal angular resolution combined with coverage of all three dimensions.

In a recent paper, I pointed out that these requirements entail a design that looks like a pair of centrioles with remarkable accuracy of the geometric details (Albrecht-Buehler, 1981). The nine concave blades appear as optimally designed blinds that restrict signal reception to a single spatial sector. The pitch of the blades permits finer angular resolution within one such sector. The well-known right-angle orientation of the two centrioles of a pair appears as a necessity to cover the third dimension in the signal detection.

If, indeed, the geometry of centrioles is the result of such engineering necessities, one could understand that nature conserved it so well. Furthermore, centrioles would appear as basic cellular instruments like ribosomes or

mitochondria, which are largely independent of any particular organism and can therefore be found in widely differing species of organisms alike.

Even if one accepts these arguments, there is still a long way to go from a signal locator to its associated “computer,” which integrates the signals according to a particular inherited logic. Yet, if one accepts the above interpretation of the centriole structure, one also accepts that a cellular data processor exists; otherwise, one would assume that cells contain an elaborate signal-detection system that has the illogical task of doing nothing more with the detected signals. I believe that intensified study of centrioles may guide us to the postulated cellular data processor, and thus to the putative core of cell behavior.

14. Outline of a Concept of Motility Control

A model of the mechanism of cytoplasmic control of cell movement seems to rest on several points:

1. Microtubules are involved in the control of cell polarity and thus in the control of the direction a migrating tissue cell takes at every turn.
2. Microtubules are focused on MTOCs around the centrioles and traverse the cytoplasm without branches to the cell edges.
3. The cell edges contain autonomous domains that can be isolated as microplasts.
4. Centrioles have a highly conserved elaborate microtubular skeleton, that can be interpreted as optimally designed for locating the source direction of unspecified signals that travel along straight lines. According to this interpretation the signal receptors are placed near each A-tubule facing the lumen of the centriole (see Fig. 1).

Combining these points, one may arrive at the outline of a model as illustrated in Fig. 1. Several steps are involved:

1. *Signaling*: Certain sources (S in Fig. 1) inside the cell's own cytoplasm and/or inside the cytoplasm of another cell emit a signal that travels along straight lines in all directions. As an illustration, one may think of infrared radiation emitted by specific vibrations of specific macromolecules (emitter molecules).
2. *Signal localization*: Receptor molecules (the small white circles in Fig. 1) are located along the inner face of the A tubule of each centriolar blade. Because of the particular design of the centriole, only one receptor can be reached by the emitted signal, as shown by the exclamation point (!) in Fig. 1. All other receptors are blocked by the blades. In this way, a one-to-one mapping between source directions and individual receptors is guaranteed. If the signal is infrared radiation, the receptor molecule could conceivably be the same as the emitter molecule and could receive the signal by resonance absorption.

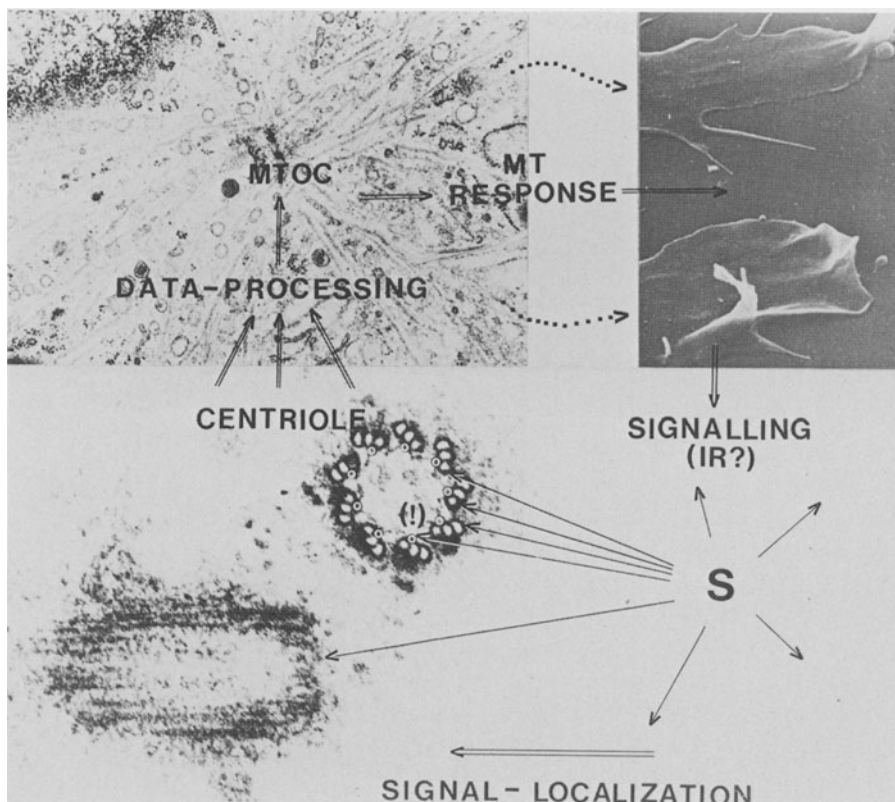


Figure 1. Schematic representation of the postulated control system of animal cell movement. A signal source S (the cell's or another cell's cytoplasm) emits an unspecified signal (possibly infrared?) that travels along straight lines in all directions. The peculiar geometry of centrioles permits it to reach the inside of only one blade (marked "!"). In this way, a one-to-one relationship between the location of the signal source and the putative signal receptors is accomplished. An unknown event of data processing is postulated to produce a change at the microtubular organizing centers (MTOC) that are associated with the centrioles. Another kind of signal is assumed to travel along particular microtubules to the peripheral domains of the cell that are able to move autonomously. The signal is assumed to influence their motile actions.

3. *Data-processing (signal integration)*: Nothing is known about this mechanism, except that it must exist, if the centriole is, indeed, a signal locator. Otherwise, the signal would be located for no purpose. This mechanism is supposed to compare, link, and integrate by certain rules of cellular logic the different incoming signals and their meanings. These rules can be expected to be genetically fixed and to play a major role in cell behavior.
4. *Efferent microtubular response*: The efferent signals that result from the data-processing mechanism must be distinguishable from the incoming signals. The processing system would otherwise process its own responses in a vicious circle. The separation between afferent and

efferent signals can be achieved by special timing, by different coding, or by sending the signals along different carriers. Considering the specific role of microtubules in the control of cell movement together with their radial arrangement with the centrioles in the center, it seems consistent to carry the concept further by assuming that the microtubules are efferent signal lines.

5. *Speculation about signal propagation along microtubules:* The microtubules, originating at the MTOCs, span without branching the space between the centriolar region and the autonomous motile domains at the cell edge.

In addition to their lack of branches, microtubules have two further properties that seem consistent with a signal propagation function. The first is that they carry a sleeve along their length. It may be interpreted as an insulating layer for the putative conduction mechanism along the tubule wall against interference by the surrounding cytoplasm (Stebbins and Willison, 1973; Albrecht-Buehler and Bushnell, 1982). The second intriguing property of microtubules in this context is the structure of their protofilaments.

Each protofilament is a chain of ($\alpha = \beta$) tubulin dimers: (α - β)(α - β)(α - β) . . . (α - β). Within the wall of a microtubule, each monomer is in contact with other monomers of the same protofilament and with those of the adjacent protofilaments. One can therefore expect that each monomer is subject to several forces of interaction between itself and at least four surrounding tubulin molecules. The total of these forces can be expected to be attractive, or else the protofilaments would not be stable, and adjacent protofilaments would separate from each other. Each of these interactions must weaken the (α - β) dimer bond. Consequently, it is conceivable that the wall of a microtubule exists in a state of resonance as to where the many kinds of intermonomeric bonds are actually located. For instance, (α - β)(α - β) . . . (α - β)(α - β) could conceivably resonate with (α)(β - α)(β . . . α)(β - α)(β). Such a resonating chain could conduct signals with almost the speed of light. For example, if a (β)-monomer is added exogenously to one end of a protofilament momentarily in the conformation (α)(β - α)(β - α) . . . (β), it could instantly shift the bond conformation. Thus, (β) + (α)(β - α)(β . . . α)(β - α)(β) could turn into the configuration (β - α)(β - α)(β . . . α)(β - α) + (β), releasing a (β) monomer at the other end.

In other words, if the cells have a mechanism whereby tubulin monomers can be added to the proximal (centriolar) end of a microtubule, it is conceivable that the distal end could release the same kind of monomers almost instantly, thereby providing a signal. The monomer released from the distal end is not the same molecule as that added at the proximal end of the protofilament. Nevertheless, as this process continues, the monomers added to the proximal end slowly and stepwise advance toward the distal end, similar to the so-called treadmilling of microtubules *in vitro* (Margolis and Wilson, 1978).

15. Dumb versus Intelligent Machines

The above outline may be taken as an illustration for the kinds of questions and approaches that result from the concept of cytoplasm intelligence. The details of the outline are testable, although it may take many years to develop the appropriate test methods.

Assume that the approach can successfully unravel the machinery of cytoplasm intelligence. Would the resulting view of the cytoplasm really be different from a concept of the cytoplasm as an automatically operating machine? There is no doubt that intelligent machines are machines, nevertheless. Only their degree of complexity seems to distinguish them from a “dumb” machine that relies merely on spring-and-lever action, yet there are important differences.

15.1. Signals versus Forces

The actions of intelligent machines are triggered by signals and are not caused by forces sufficient to produce the action. For example, a person ordered out of a room is not removed by the mechanical pressure of the sound waves of the words. They only trigger the person’s control systems of walking.

15.2. Ambiguity of Response

Another important difference between intelligent and dumb machines lies in the ambiguity of their responses. A spring-and-lever construction has only one built-in response for each type of input. For example, a mechanical watch can only turn a wheel in response to winding it up, whereas a computer can respond to a key stroke in a multiplicity of ways determined by the presence or absence of results of previous calculations, input data, or the simultaneous action of peripherals. A similar ambiguity can be observed among the chemicals of the cytoplasm as opposed to the single outcome of other chemical reactions. For example, the reaction between hydrogen and oxygen has only one explosive outcome. In contrast, α - and β -tubulin in cells can remain as dimers, can assemble single microtubules, form spindles, centrioles, the tube of feeding tentacles, the cell wall of trypanosomes, or many more elaborate structures.

15.3. Accuracy and Adaptability

Compare a cannon with a heat-seeking missile. The cannon has to be pointed with utmost accuracy at the target while anticipating wind, air temperature, visibility, and movement of the target. On the other hand, the heat-seeking missile that is computer-controlled during its flight needs only to

be pointed in the approximate direction of the target. From thereon, it will adapt to the changing atmospheric conditions and aim at the target with greater accuracy, the closer it comes.

15.4. Expandability to Higher Degrees of Complexity

Perhaps the most important difference between an intelligent and a dumb machine is the ability of intelligent machines to form larger networks of even more complex intelligent machines. A box full of mechanical watches will never turn into a time keeper of an astronomical observatory. A network of linked computers, however, can take on more and more complex control tasks. Owing to the speed of the exchanged signals and their weak, i.e., non-damaging impacts on the machine itself, very large numbers of them can be simultaneously sent between the intelligent components of the larger network, resulting in the establishment of a higher stratum of organization of the intelligent components. Perhaps, tissue formation from "intelligent" cells can be seen from this angle.

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